



RAKIB HASAN

Data Analyst | Data Scientist | MSc Business Data Science

Machine Learning & AI | NLP/LLMs | MLOps Foundations

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[LinkedIn](#) • [Portfolio](#) • [GitHub](#)

PROFILE

Recent MSc Business Data Science graduate from Aalborg University, June 2026, with five years of prior operational and data management experience, now backed by hands-on training in machine learning, NLP, and LLM-based analysis. Built and deployed a full-stack AI travel-planning platform for Denmark, combining a RAG pipeline, a validated multi-agent LLM system with deterministic backend routing, and a trained ML recommendation model, live end-to-end across a FastAPI backend, PostgreSQL/pgvector database, and Next.js/React frontend. Comfortable moving between high-level analysis and hands-on technical execution, with a track record of turning messy operational data into decision-ready reporting that management relied on daily.

TECHNICAL SKILLS

- **Programming & Data:** Python (Pandas, NumPy), SQL, Git/GitHub version control, data cleaning & preprocessing, feature engineering, ETL/ELT data pipelines, data governance, data quality, data validation, A/B testing, FastAPI, REST API design, PostgreSQL (Neon, pgvector, vector search)
- **Machine Learning & AI:** Scikit-learn (regression, classification, time-series forecasting, evaluation metrics), statistical analysis, MLflow, Optuna, XGBoost, LightGBM, PyTorch, TensorFlow, Docker, NLP, LLM workflows (prompt engineering, RAG, LangChain, LiteLLM, Ollama, Groq, OpenAI API), agentic systems (CrewAI, multi-agent orchestration), structured output (Pydantic, instructor), embeddings (sentence-transformers, fastembed), Hugging Face Transformers (DistilBERT)
- **Frontend:** React, Next.js (App Router, Server & Client Components), TypeScript, Tailwind CSS, Recharts
- **Visualisation & Reporting:** Matplotlib, Seaborn, Streamlit dashboards, Microsoft Excel (pivot tables), management reporting, business intelligence, KPI reporting, stakeholder management
- **Data Collection & Automation:** Requests, BeautifulSoup, Playwright, trafilatura, web scraping, geo point-in-polygon matching (Shapely, osmium), Nominatim, Serper API, pytest, ruff linting, GitHub Actions (CI & scheduled workflows), cloud deployment (Render, Vercel)
- **Languages:** English (Fluent), Danish (Beginner)

SELECTED PROJECTS

AI Denmark Explorer | *Personal Project*

2026

- Built and deployed a full-stack place-discovery platform for Copenhagen, combining a Next.js/React frontend (Vercel), a FastAPI backend (Render), and a Neon PostgreSQL/pgvector database.
- Engineered a RAG pipeline with semantic search (fastembed embeddings, pgvector similarity search) over real Copenhagen places sourced from OpenStreetMap, grounding GPT-4o summaries in place data, with a live Nominatim/Serper fallback for thin data, validated via a live A/B test on prompt quality.
- Architected a two-agent CrewAI system (OpenAI GPT-4o-mini, GPT-4o) with Pydantic/instructor-validated structured intent, deterministically routed (near/far/sequential/area) instead of free-form AI tool calls, grounded in live Open-Meteo weather.
- Built and backtested a live XGBoost recommendation-confidence classifier (Optuna-tuned, MLflow-tracked) combining offline DistilBERT sentiment with real-time semantic signals, lifting correlation with real outcomes from Pearson $r = 0.171$ to $r = 0.713$ against the prior quality-score model.

[Live app](#) [GitHub](#)

AI-Assisted Media Bias Detection in Geopolitical News | *Master's Thesis*

2025 - 2026

- Built and shipped an end-to-end Python NLP/LLM pipeline that collects, cleans, and analyses geopolitical news coverage across four major international outlets (BBC, The Guardian, Daily Star, New Age).

- Grew and maintained a continuously updating corpus of 6,500+ articles in a SQLite database, with automated deduplication, mixed-date parsing, and topic classification across heterogeneous sources, refreshed daily via GitHub Actions.
- Combined TF-IDF retrieval with GPT-4o analysis to score framing, selection/omission, sourcing, and tonal bias, surfacing findings through an interactive Streamlit app.

[Live app](#) [GitHub](#)

EDUCATION

MSc in Business Data Science | Aalborg University, Denmark

Sep 2024 - Jun 2026

- Master's thesis: AI-assisted media bias detection using NLP, LLM-based analysis, and a SQL (SQLite) backend (see Selected Projects).
- Key coursework: applied machine learning, NLP & network analysis, deep learning & AI, data-driven business modelling, data governance & ethics, data engineering & MLOps.

Bachelor of Business Administration (BBA) | University of Dhaka

Jan 2014 - Sep 2018

- **Major:** Management Information Systems
- Coursework: data-driven decision-making, information systems, business strategy, and leadership.

PROFESSIONAL EXPERIENCE

Assistant Manager (Data & Reporting) | V. A. Transport Agency

Jan 2022 - Jun 2024

- Owned the company's operational databases end-to-end, safeguarding the accuracy and consistency of the data behind every team's reporting.
- Built and maintained the recurring dashboards and reports that became management's standard basis for day-to-day logistics decisions, replacing ad hoc, manually assembled updates.
- Analysed operational data to surface inefficiencies and partnered with cross-functional teams to streamline scheduling and workflows, reducing recurring planning errors and delays.

Senior Officer | V. A. Transport Agency

Mar 2019 - Dec 2021

- Maintained operational records and service data, coordinated transport operations and customer communication, and drove process improvements that made reporting faster and more consistent.

Part-Time Sushi Chef | Karma Sushi Aalborg & Catch Sushi Bar

Jan 2025 - Jun 2026

- Balanced kitchen work with full-time studies and thesis research, maintaining quality control and composure under pressure while coordinating with and onboarding team members during high-volume service.

ADDITIONAL PROJECTS

- **Salary Analysis & Tax Calculator:** Streamlit app for gross/net income, tax deductions, and monthly earnings tracking (Python, Pandas). [Live app](#)
- **CO2 Emissions Forecasting (Energy Sector):** group MLOps project forecasting energy-sector emissions (Python, Scikit-learn, Streamlit).
- **Customer Service Chatbot:** NLP chatbot for order status, payment, and refund support (Python, Ollama, LangChain, SentenceTransformers, Streamlit).
- **Restaurant Guest Forecasting (Karma Sushi Aalborg):** forecasting models predicting daily guest numbers to inform staffing decisions (Python, Pandas, Scikit-learn). [GitHub](#)

VOLUNTEERING & LEADERSHIP

- **Volunteer:** Cross-Cultural Trainers (CCT) & International Cultural Youth Exchange (ICYE), Denmark: Community and intercultural engagement (2025 - present)
- **Buddy, International Buddy Program:** Aalborg University, supported international students adapting to university life, Sep 2025 - Feb 2026. [Certificate](#)
- **Student leadership:** General Secretary, MIS Hub 71; Organizing Secretary, MIS Debating Club; Head, MIS Cultural Wing.